

NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations

December 2023

ME1008-1 – INTRODUCTION TO INTERNET OF THINGS
(ME)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. **What does networking refer to?**
 - A) Linking computers and communication devices
 - B) Developing software applications
 - C) Manufacturing computer hardware
 - D) Analyzing data patterns
2. **What types of data are commonly transferred between hosts in a network?**
 - A) Text and audio
 - B) Images and videos
 - C) Analog signals
 - D) Physical documents
3. **What types of systems are point-to-point connections functional for?**
 - A) Only synchronous systems
 - B) Only asynchronous systems
 - C) Both synchronous and asynchronous systems
 - D) None of the above
4. **What does network topology refer to in the context of computer networks?**
 - A) The study of data transmission
 - B) The arrangement of links and nodes in a computer network
 - C) The speed of data transfer
 - D) The type of data being transmitted
5. **How many dedicated full duplex links are there between n hosts in a mesh network?**
 - A) n links
 - B) $n(n-1)$ links
 - C) $2n$ links
 - D) $(n-1)$ links
6. **In the context of IoT applications, what is the first step in a majority of scenarios?**
 - A) Data transmission
 - B) Data analysis
 - C) Sensing
 - D) Actuation
7. **Which term describes the degree of closeness between a sensor's measured value and the true or actual value of the quantity being measured?**
 - A) Sensing range
 - B) Accuracy
 - C) Precision
 - D) Sensitivity
8. **Why is energy efficiency important in sensor design and operation?**
 - A) To make the sensor physically smaller
 - B) To reduce the sensing range
 - C) To extend the sensor's battery life or reduce power consumption
 - D) To increase the sensor's precision
9. **What is the role of languages like Structured Query Language (SQL) in dealing with structured data?**
 - A) They are used for generating random data
 - B) They are used for analyzing unstructured text
 - C) They are used for accessing and querying structured data in RDBMS
 - D) They are used for processing image data
10. **What are the three types of data classification based on the urgency of processing in the context of IoT?**
 - A) Immediate, fast, and slow
 - B) High, medium, and low
 - C) Very time critical, time critical, and normal
 - D) Critical, urgent, and non-urgent

11. What is the primary advantage of on-site processing in IoT applications, as described in the content?
 - A) Reduced data generation rate
 - B) Greater tolerance for latencies
 - C) Enhanced data storage capacity
 - D) Fast and robust data processing
12. Which approach for offloading data in an offsite-processing IoT deployment involves maximizing certain QoS parameters while reducing the qualities of others?
 - A) Naive Approach
 - B) Bargaining based approach
 - C) Learning based approach
 - D) Statistical approach
13. How does virtualization make resources appear to end users?
 - A) Limited and dedicated
 - B) Scarce and costly
 - C) Unlimited and dedicated
 - D) Unavailable and shared
14. What does SLA stand for in the context of cloud computing?
 - A) Software Licensing Agreement
 - B) Service-Level Agreement
 - C) Storage Location Agreement
 - D) System Logging Agreement
15. Why is the "Penalty" metric included in an SLA?
 - A) To specify the service's availability duration
 - B) To define the problem reporting process
 - C) To measure the flexibility of data transfer
 - D) To provide consequences for not meeting SLA promises
16. Which feature of an IoT device/node determines its capabilities, including local data processing, data storage, and filtering?
 - A) Form factor
 - B) Energy source
 - C) Memory capacity
 - D) Processing power
17. What are the main components of machine learning that play a crucial role in designing intelligent systems in IoT?
 - A) Electronics, chemistry, and biology
 - B) Statistics, mathematics, and computer science
 - C) History, literature, and philosophy
 - D) Physics, engineering, and astronomy
18. Which category of machine learning algorithms does KNN learning belong to?
 - A) Unsupervised learning
 - B) Reinforcement learning
 - C) Supervised learning
 - D) Semi-supervised learning
19. What distance metric is commonly used in k-means clustering to measure the similarity between data points and cluster centroids?
 - A) Pearson correlation coefficient
 - B) Cosine similarity
 - C) Euclidean distance
 - D) Jaccard similarity
20. What is one of the significant challenges in machine learning related to data quality?
 - A) Having an insufficient amount of data
 - B) Selecting the right machine learning model
 - C) Ensuring the data is informative and meaningful
 - D) Using noisy or erroneous data for training

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	List and explain the different types of physical topologies in Computer Network.	8	L*2	1	1
b)	What is the primary function of an actuator in a machine or system? How does the control system interact with the actuator to achieve its desired outcome?	8	L2	2	1
2. a)	Define IoT. Explain the evolution of IoT.	8	L2	1	1
b)	How are various sensors classified? Explain it in detail.	8	L2	2	1
3. a)	Present the advantages and disadvantages of different topologies.	8	L2	1	1
b)	Explain the Functional blocks of a typical sensor node in IoT.	8	L2	2	1
Unit – II					
4. a)	What are the different data formats found in IoT network traffic streams? Explain	8	L2	3	1
b)	What is an SLA? Why it is important in cloud computing?	8	L2	4	1

5.	a)	How is collaborative processing different from remote processing?	8	L2	3	1
	b)	What is a sensor-cloud? Why do we use sensor-cloud?	8	L2	4	1
6.	a)	What are the various decision making approaches chosen for offloading data in IoT?	8	L2	3	1
	b)	Design a case study to develop an IoT-based agricultural planter. In the case study, you should include the requirement analysis of different components and justify their usability in the planter.	8	L4	4	1

Unit – III

7.	a)	What are the key components of the Internet of Things (IoT) in healthcare, and how do they contribute to improving patient care and healthcare operations?	8	L2	5	1
	b)	What is Machine Learning? What are the advantages and challenges associated with the application of machine learning in various industries.	8	L2	5	1
8.	a)	What is the Random Forest machine learning algorithm? Explain it with an example of how it works in practice.	8	L2	5	1
	b)	What is the k-means clustering algorithm, and outline the key steps involved in its application for data analysis and pattern recognition?	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
